Fiscal Year:	FY 2023	**Task Last Updated:**	FY 07/31/2022
PI Name:	Zea, Luis Ph.D.		
Project Title:	Characterization of Clinically- and Spaceflight-Relevant Bacteria and Fungi and Drug Effectiveness under Simulated Lunar, Martian, and Microgravities		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Boulder	**State:**	CO
Zip Code:	80309-0429	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	10/01/2018	**End Date:**	12/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:	Freeland, Denise	**Contact Phone:**	321-867-5878
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2021 per NSSC information (Ed., 7/12/21) NOTE: End date changed to 3/31/2021 per NSSC information (Ed., 9/10/20) NOTE: End date changed to 9/30/2020 per NSSC information (Ed., 7/8/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Stodieck, Louis Ph.D. (University of Colorado, Boulder)		
Grant/Contract No.:	80NSSC18K1468		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Microbial (bacteria and fungi) behavior has been observed to change in microgravity: enhanced bacterial growth, improved cell aggregation, increased virulence, and reduced susceptibility to antibiotics have been reported on spaceflight experiments. However, these parameters have not been systematically assessed, and even less so at partial gravities: 1/6 g (lunar) and 1/3 g (Mars). This project aims to characterize changes on cell growth dynamics and cell aggregation as a function of these gravitational regimes (simulated micro-, 1/6, and 1/3 g). This will be complemented by gene expression analyses focused on the drug-resistance-associated genes (resistome) as well as those responsible for bacterial virulence. This systematic microbial study will include ten commensal, opportunistic, and pathogenic bacterial and fungal strains (of Escherichia coli, Pseudomonas aeruginosa, methicillin-resistant Staphylococcus aureus (MRSA), Candida albicans, Bacillus cereus, Staphylococcus haemolyticus, Salmonella typhimurium, and Shewanella oneidensis). This study has a second component that focuses on determining if there are changes on the concentrations of drugs needed to inhibit microbial growth at different gravitational regimes, like the ones expected to be experienced during deep space missions, the Moon, and Mars. For this study, four different antimicrobials (each representing distinct mechanisms of action) are planned to be used: gentamicin sulfate (aminoglycoside class), which has been used for three different space-based experiments; ciprofloxacin (fluoroquinolone class), the most commonly prescribed drug for urinary tract infections (UTI); Ceftolozane/Tazobactam (ß-lactam/ ß-lactam combination antibiotic used on drug-resistant infections); and Fluconazole (an azole-class antifungal). The use of spaceflight hardware, inclined clinostats to simulate partial- and microgravity, and human microbiome-associated bacterial and fungal models, will enable this project to address Risks and Gaps of NASA's Human Research Roadmap, as well as to expand our understanding of fundamental cellular and molecular mechanisms behind microbial responses to changes in gravity. Knowledge produced from this project may help us better prepare for human space exploration of other celestial bodies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying gene expression of bacteria and fungi exposed to antimicrobials under altered gravitational conditions may help unmask molecular genetic processes not observable under the 1 g environment on Earth. In turn, this may help scientists learn of how bacteria are able to develop resistance mechanisms and/or potentially find novel targets against drug-resistant bacteria.
Task Progress:	This project aimed to characterize changes on cell growth dynamics and cell aggregation as a function of gravitational regime (simulated micro-, 1/6 and 1/3 g). This was complemented by gene expression analyses focused on the drug-resistance-associated genes (resistome), as well as those responsible for bacterial virulence. This systematic microbial study included multiple commensal, opportunistic, and pathogenic bacterial and fungal strains. The study had a second component that focused on determining if there were changes in the concentrations of drugs needed to inhibit microbial growth at different gravitational regimes. Ten different strains of bacteria (three strains of Escherichia coli, Pseudomonas aeruginosa, methicillin-resistant Staphylococcus aureus (MRSA), Salmonella, Staphilococcus epidermidis, and Shewanella oneidensis) and of fungi (Saccharomyces cerevisiae and Candida albicans) were investigated in this project. They were cultured separately in BioServe's Fluid Processing Apparatus (FPA), which have been operated over 5,000 times in orbit on more than 40 spaceflight experiments. This helped our team compare clinostat-produce data against sets derived from actual microgravity. Reduced gravitational conditions were achieved using BioServe's FPA Clinostats. The project has now concluded. Two papers to be submitted for open-access journal publication, plus an additional conference paper, are currently in progress.
Bibliography Type:	Description: (Last Updated: 11/13/2025)
Articles in Peer-reviewed Journals	Gumulya Y, Zea L, Kaksonen AH. "In situ resource utilisation: The potential for space biomining." Minerals Engineering. 2022 Jan 6;176:107288. Review. https://doi.org/10.1016/j.mineng.2021.107288 , Jan-2022
Articles in Peer-reviewed Journals	Santomarino R, Zea L, Cockell CS. "The smallest space miners: principles of space biomining." Extremophiles. 2022 Jan 6;26:7. https://doi.org/10.1007/s00792-021-01253-w , Jan-2022
Articles in Peer-reviewed Journals	Allen LA, Kalani AH, Estante F, Rosengren AJ, Stodieck L, Klaus D, Zea L. "Simulated micro-, lunar, and Martian gravities on Earth—effects on Escherichia coli growth, phenotype, and sensitivity to antibiotics." Life. 2022 Sep 8;12(9):1399. https://doi.org/10.3390/life12091399 , Sep-2022
Books/Book Chapters	Pathak, Y, Araújo dos Santos M, Zea L. (Eds.) "Handbook of Space Pharmaceuticals." Cham, Switerland: Springer, 2022. 978-3-31-950909-9. https://doi.org/10.1007/978-3-030-05526-4 , Jun-2022